MONASH University

FIT5196 DATA WRANGLING

Week 2

Data Wrangling Process & Tasks

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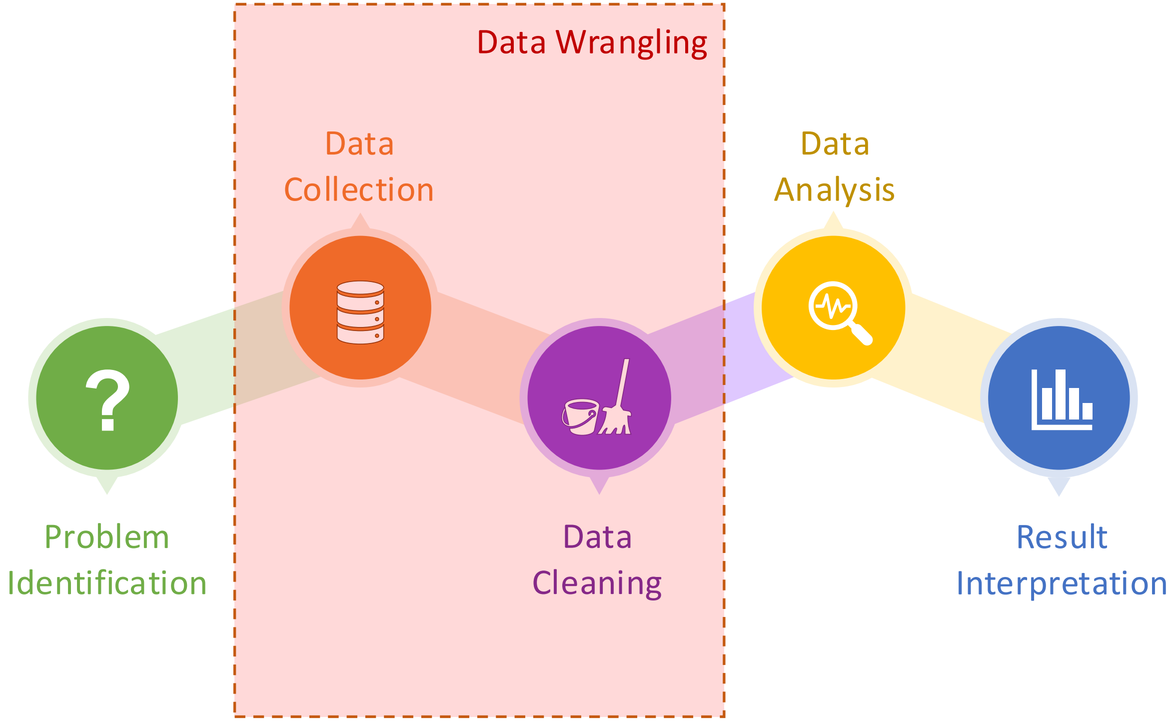
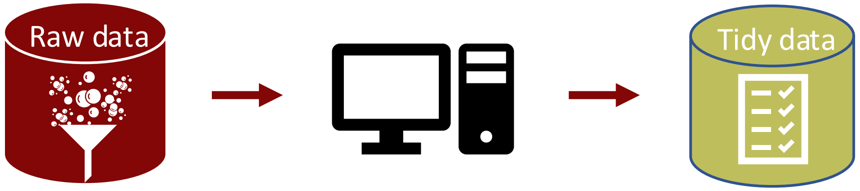
Data Wrangling

- **Data Wrangling** is the process of acquiring, cleaning, structuring, and enriching raw data into a format that is directly usable for analysis.



Goals of Data Wrangling

- The goals of data wrangling are multifaceted, aiming to simplify data analysis and maximize the value extracted from the data.
  - Improving Data Quality
  - Data Formatting and Standardization
  - Simplifying Access to Data
  - Enriching Data
  - Reducing Data Complexity
  - Facilitating Data Integration
  - Increasing Analytical Efficiency
  - Supporting Decision Making



Data + Wrangling + Analysis
= Data Product (Knowledge)

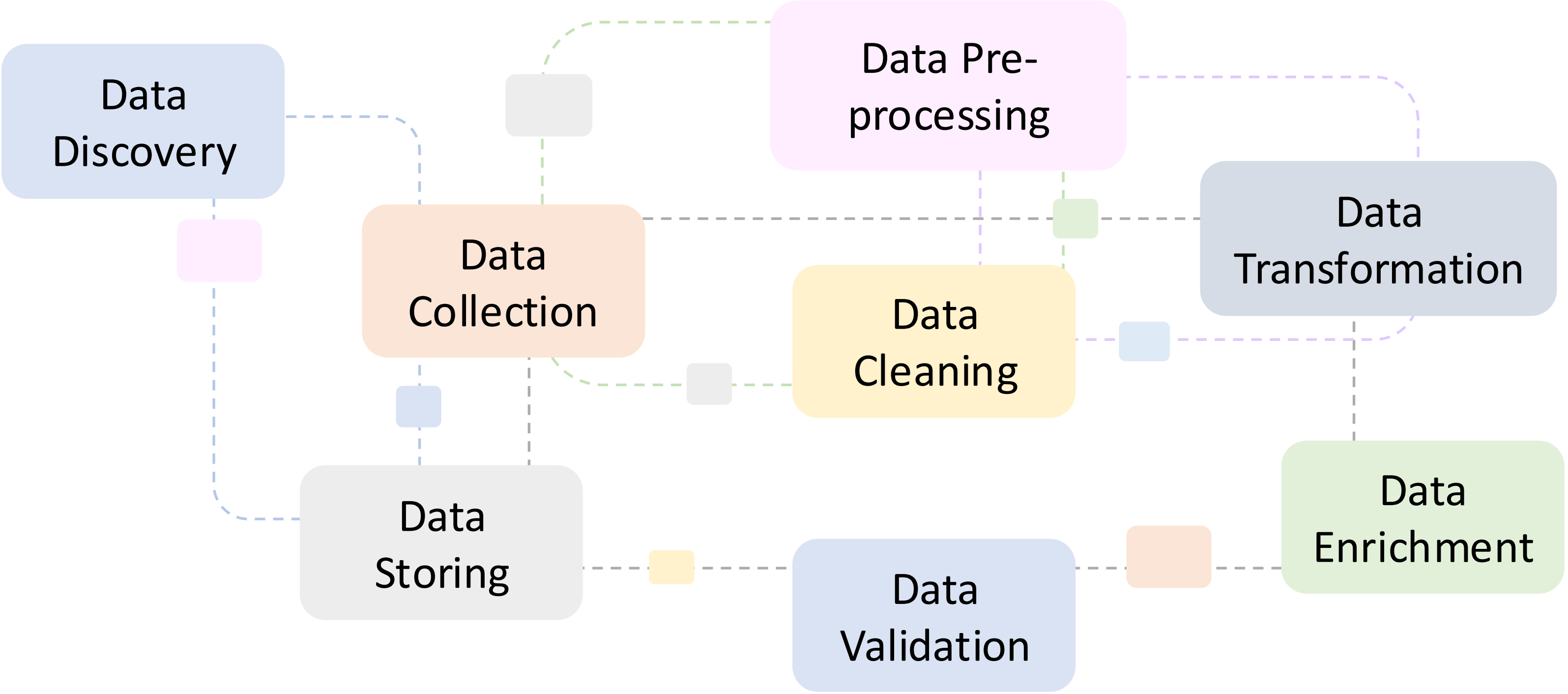
Challenges in Data Wrangling

- Challenges arise from the nature of the data itself, the complexity of data sources, and the goals of the data analysis projects.
    - Volume of Data & Scalability
    - Data Quality Issues
    - Data from Diverse Sources
    - Complexity of Data Structures
    - Lack of Standardization & Interpretability
    - Highly Time-Consuming
    - Skill and Tool Requirements
    - Data Privacy and Security

Andrew Kamau, https://blog.google/products/chrome/5-tips-to-stay-safer-online-with-chrome/

Data Wrangling Tasks

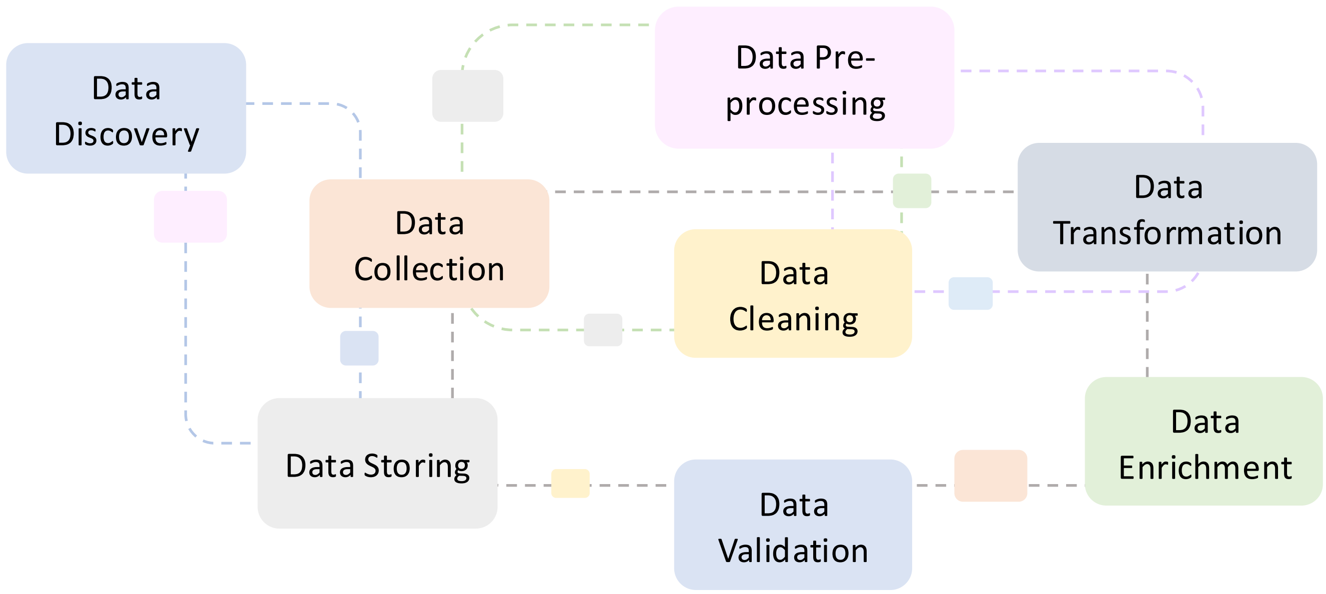
- **Data Wrangling** is the process of acquiring, cleaning, structuring, and enriching raw data into a format that is directly usable for analysis.



Outline

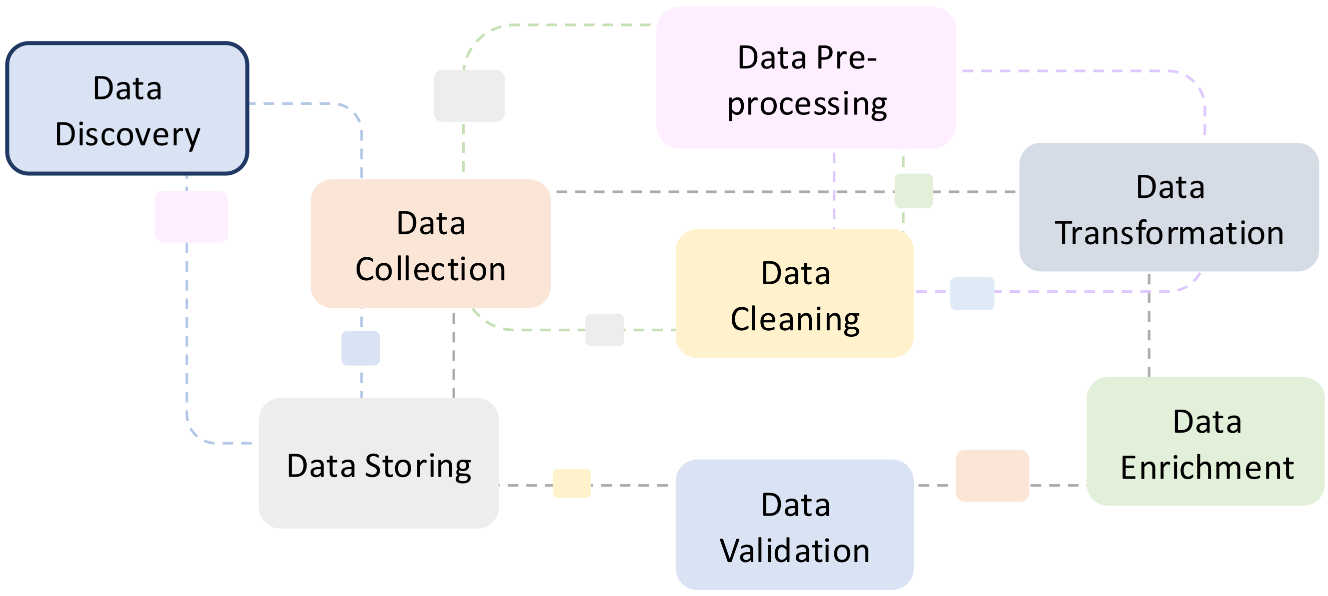
- Data Wrangling Process
- Main Tasks in Data Wrangling
  - Data Discovery
  - Data Collection
  - Data Pre-processing
  - Data Cleaning
  - Data Transformation
  - Data Enrichment
  - Data Structuring
  - Data Validation

Overview of Data Wrangling Process



Data wrangling is the process of making data useful.

Overview of Data Wrangling Process



Data Discovery

The initial step where data scientists identify and access various data sources relevant to the analytical objectives.

This includes determining the availability, suitability, and quality of data.

Data wrangling is the process of making data useful.

Example – Data Discovery

Project: **Predictive Analysis for Patient Readmissions**

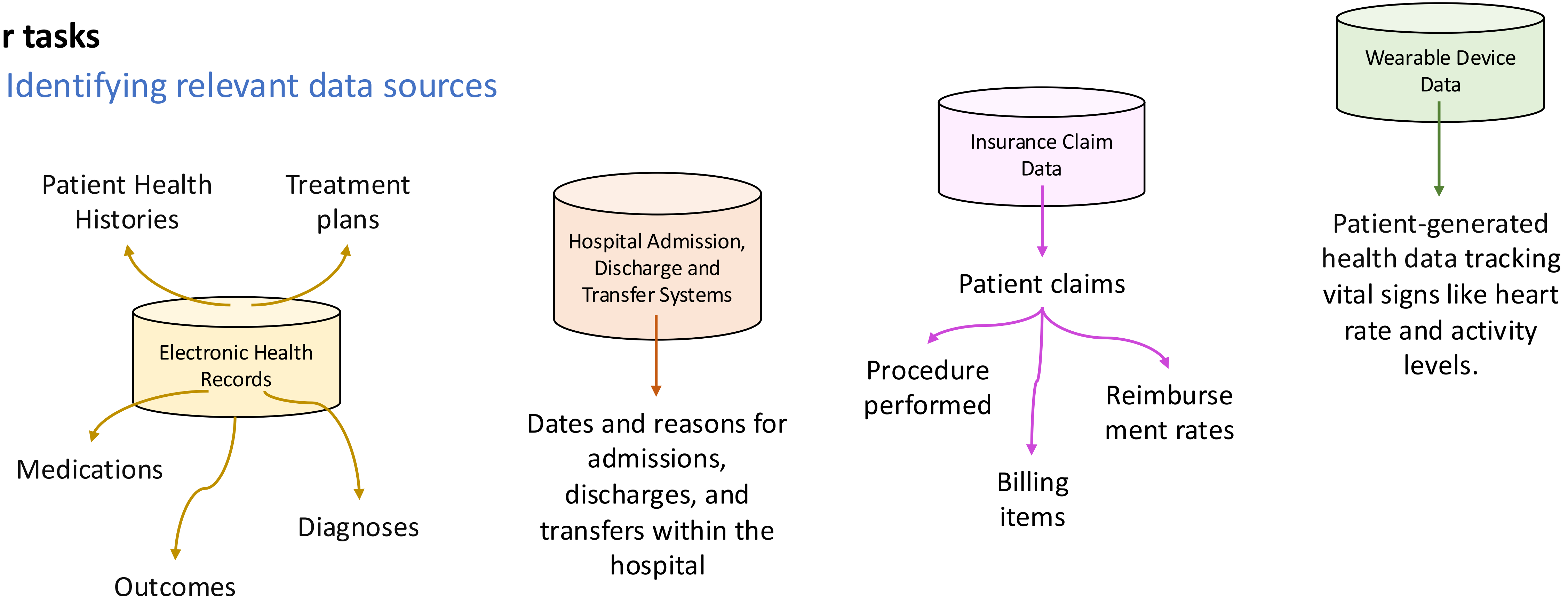
- **Objective –** Aims to develop a predictive model to identify patients at high risk of readmission within 30 days of discharge
- **Goal –** To improve patient outcomes and reduce healthcare cost
- **Your tasks** in a healthcare analytics team
  - Identifying relevant data sources
  - Understanding data availability and accessibility
  - Assessing data quality and suitability

John Russell, 2021, Hospitals face big fines for frequent readmission, https://www.ibj.com/articles/hospitals-face-big-fines-for-frequent-readmissions

Example – Data Discovery (cont.)

Your tasks

- Identifying relevant data sources



Example – Data Discovery (cont.)

Your tasks

- Identifying relevant data sources
- Understanding data availability and accessibility

Insurance Companies

Hospital IT Department

Doctors, Nurses, other Clinicians or Health worker

Health Monitoring Programs

Example – Data Discovery (cont.)

Your tasks

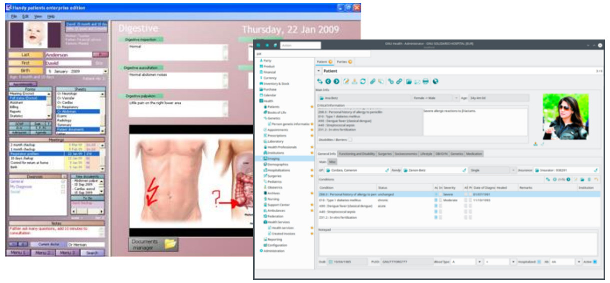
- Identifying relevant data sources
- Understanding data availability and accessibility
- Assessing data quality and suitability

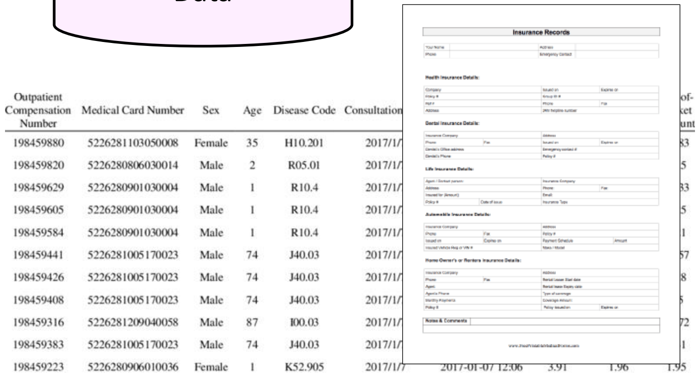
Electronic Health Records

Hospital Admission, Discharge and Transfer Systems

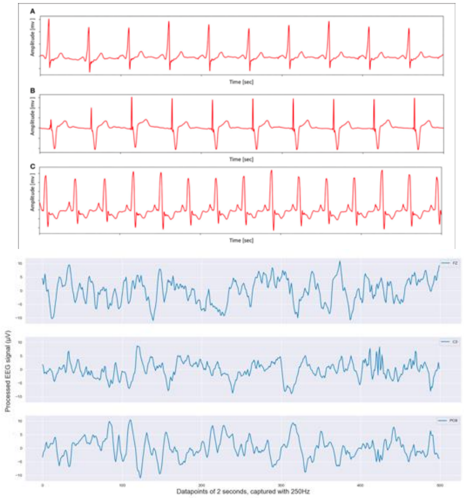
Insurance Claim Data

Wearable Device Data





Outpatient Compensation Number	Medical Card Number	Sex	Age	Disease Code	Consultation
198459880	5226281103050008	Female	35	H10.201	2017/1/
198459820	5226280806030014	Male	2	R05.01	2017/1/
198459629	5226280901030004	Male	1	R10.4	2017/1/
198459605	5226280901030004	Male	1	R10.4	2017/1/
198459584	5226280901030004	Male	1	R10.4	2017/1/
198459441	5226281005170023	Male	74	J40.03	2017/1/
198459426	5226281005170023	Male	74	J40.03	2017/1/
198459408	5226281005170023	Male	74	J40.03	2017/1/
198459316	5226281209040058	Male	87	I00.03	2017/1/
198459383	5226281005170023	Male	74	J40.03	2017/1/
198459223	5226280906010036	Female	1	K52.905	2017/1/



Completeness

Accuracy

Formatting

Structuring

Multi-data sources

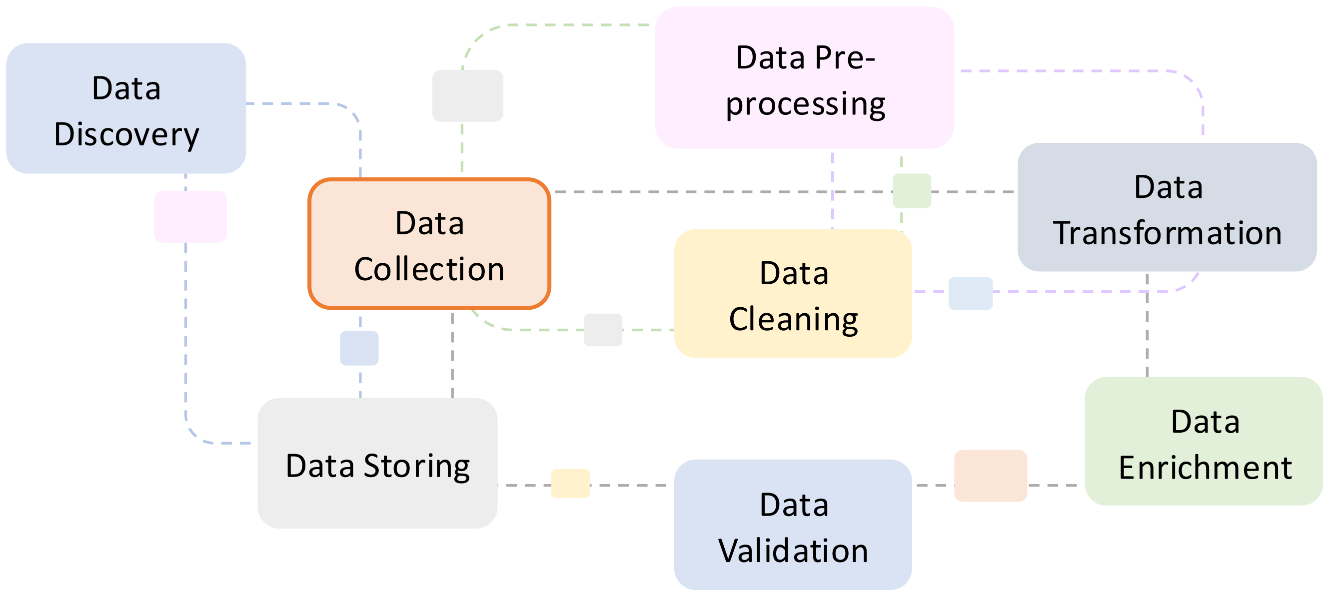
Quality

Coverage

Electronic health record, Wikipedia, https://en.wikipedia.org/wiki/Electronic_health_record

Zhou, Shengyao & He, Jie & Yang, Hui & Chen, Donghua & Zhang, Runtong. (2020). Big Data-Driven Abnormal Behavior Detection in Healthcare Based on Association Rules. IEEE Access. PP. 1-1. 10.1109/ACCESS.2020.3009006.

Overview of Data Wrangling Process



Data Collection

This phase involves gathering the identified data from its sources.

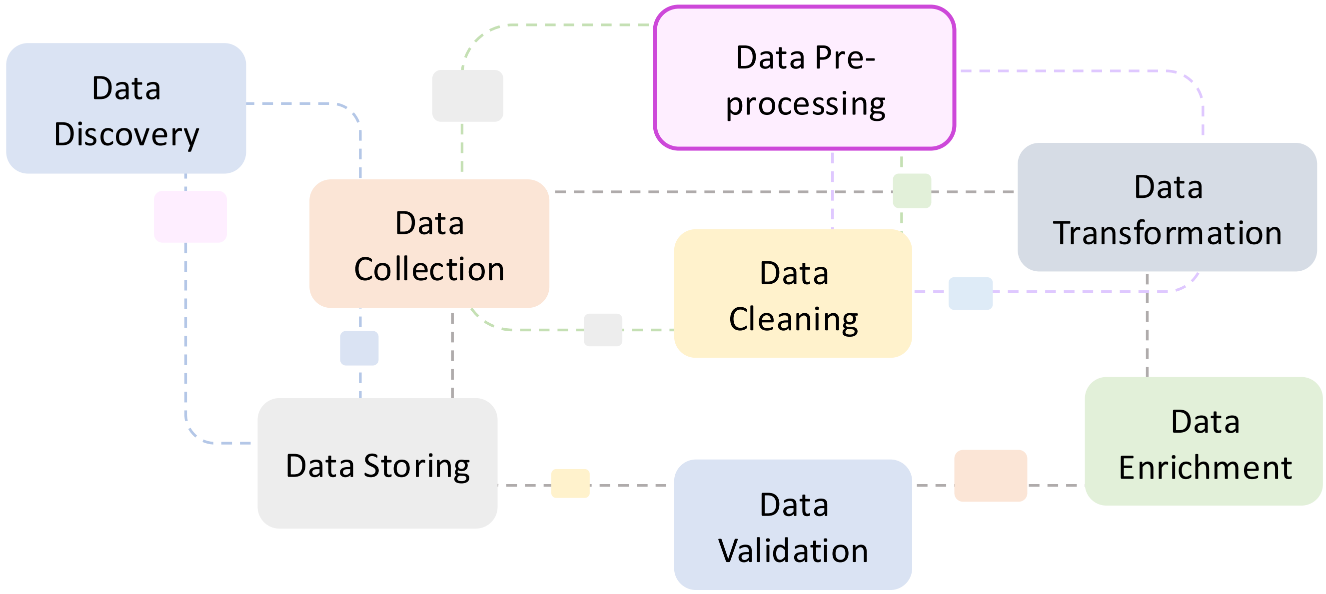
The data can be collected from multiple sources like databases, spreadsheets, APIs, and external datasets.

Data wrangling is the process of making data useful.

Data Collection

- To collect data that is relevant, accurate and robust for data analysis needs, we need to consider
  - **Define objective** – what to achieve with the data collection?
  - **Data requirements** – what the specific data needed to meet the objective?
  - **Data sources** – where to obtain the data?
  - **Data quality** – ensure the data collected is of high quality
  - **Data volume** – assess the volume of data needed and the capacity of handling
  - **Ethics** – ensure ethical practices complying with legal standards and permissions
  - **Data format** – what format to be used?
  - **Collection methods and tools** – choose appropriate methods and right tools
  - **Timeframe and budget** – align with project deadlines and budget constraints
  - **Data privacy and security** – protect privacy from individuals and secure storage and handling of data
  - **Documentation** – keep detailed documentation of the data collection process
  - **Pilot test** – validate data collection process and instruments
  - **Review and adaptation** – regularly review the process for any necessary adjustments

Overview of Data Wrangling Process



Data Pre-processing

At this stage, preliminary data preparation tasks are performed to make raw data more suitable for analysis.

It involves selecting, and formatting datasets to ensure they are in a usable state.

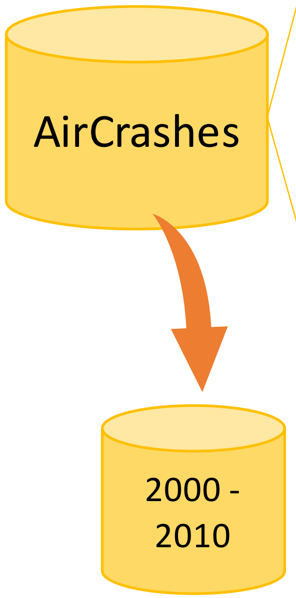
Data wrangling is the process of making data useful.

Data Pre-Processing

- **Selection**
  - **Subsetting**
    - Choosing a subset of data that is relevant to the analysis, which may involve filtering rows or columns based on certain criteria.

Example - Subsetting

Objective: To analyse accidents happened from 2000 to 2010.



- Airline_flight
- Casualties
- Circumstances
- Country
- Crew
- Data
- Ground
- Latitude
- Longitude
- Passengers
- Dead

```python
import pandas as pd
# assuming 'AirCrashes.csv' is a CSV file containing dataset
df = pd.read_csv("AirCrashes.csv")
df.head()
```

Aeroflot Flight 217	**Ilyushin Il-62**	Extremely High	Bad Visibility by Day	Russia	10	1972-10-13	0	55.755826	Moscow - Russia	37.6173	No survivors	164	APR	Accident	174	COM
Aeroflot Flight 3352	**Tupolev Tu-154**	Extremely High	Bad Visibility by Night	Russia	5	1984-10-11	4	54.9666667	Omsk - Russia	73.3833333	Some survivors	169	LDG	Accident	178	COM
										628	No survivors	156	ENR	Accident	166	COM
										556	No survivors	191	ENR	Accident	0	COM
										111	No survivors	165	ENR	Accident	178	COM

```python
# subsetting by criteria
start_date = '2000-01-01'
end_date = '2010-12-31'

df_subset = data[(data['Date'] >= start_date) & (data['Date'] <= end_date)]

df_subset
```

Air France Flight 447	**Airbus A330-203**	Extremely High	Bad Visibility by Night	Sao Pedro	12	2009-06-01	0	03.065833	Atlantic Ocean - Sao Pedro	-30.561667	No survivors	216	ENR	Accident	228	COM
American Airlines Flight 11	**Boeing 767-223ER**	Extremely High	Good Visibility by Day	US	11	2001-09-11	1600	40.7143528	New York - New York - US	-74.0059731	No survivors	81	ENR	Attack	1692	INH
American Airlines Flight 587	**Airbus A300B4-605R**	Extremely High	Bad Visibility by Night	US	9	2001-11-12	5	40.5759385	Belle Harbor - New York - US	-73.8481906	No survivors	251	ENR	Accident	265	COM
American Airlines Flight 77	**Boeing 757-223**	Extremely High	Bad Visibility by Day	US	6	2001-09-11	125	38.8799697	Arlington - Virginia - US	-77.1067698	No survivors	64	ENR	Attack	189	INH
Caspian Airlines Flight 7908	**Tupolev Tu-154M**	Extremely High	Bad Visibility by Night	Iran	12	2009-07-15	0	36.266819	Qazvin - Iran	50.003811	No survivors	156	ENR	Accident	168	COM
China Airlines Flight 611	**Boeing 747-209B**	Extremely High	Bad Visibility by Night	China	19	2002-05-25	0	23.5025	Taiwan Strait - Penghu Islands - China	119.5052778	No survivors	206	ENR	Accident	225	COM
Iranian Air Force (15-2280)	**Ilyushin Il-76MD**	Extremely High	Bad Visibility by Night	Iran	18	2003-02-19	0	30.28027	Kerman - Iran	57.06702	No survivors	257	ENR	Accident	275	MIL
Kenya Airways Flight 431	**Airbus A310-304**	Extremely High	Bad Visibility by Night	Cote d'Ivoire	10	2000-01-30	0	3.7355442	Gulf of Guinea - Cote d'Ivoire	3.743509	Some survivors	159	ENR	Accident	169	COM
Pulkovo Flight 612	**Tupolev Tu-154M**	Extremely High	Bad Visibility by Night	Ukraine	10	2006-08-22	0	49.0702778	Novoselivka - Donetsk Obl. - Ukraine	37.6913889	No survivors	160	ENR	Accident	170	COM

Data Pre-Processing

- **Selection**
  - Subsetting
    - Choosing a subset of data that is relevant to the analysis, which may involve filtering rows or columns based on certain criteria.
  - **Sampling**
    - If the dataset is too large to process or if you want to perform initial exploratory analysis, you might draw a representative sample of the data.

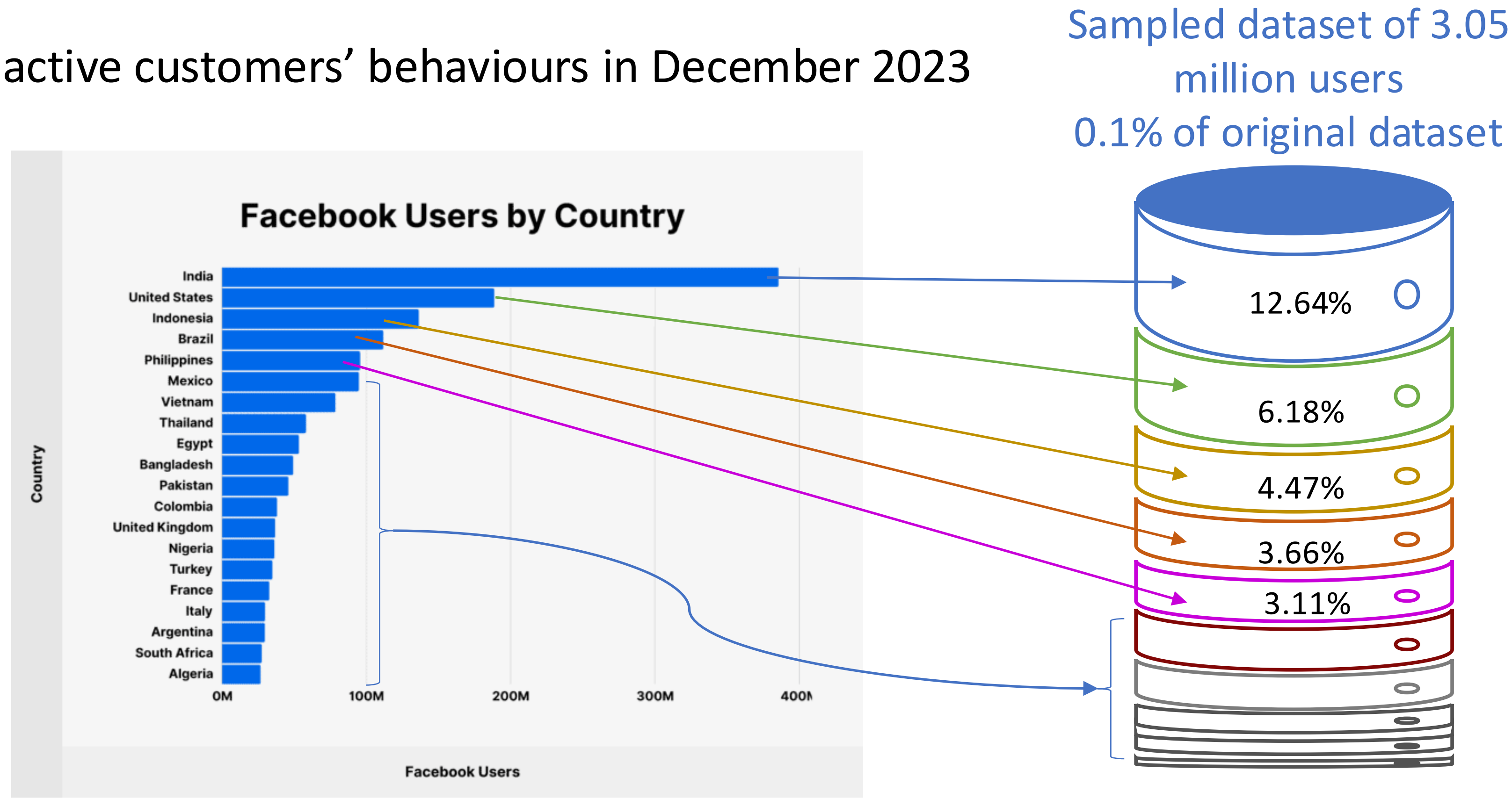
Example - Sampling

- Objective: To analyse Facebook active customers' behaviours in December 2023

Facebook monthly active users:

3.05 billion monthly users access Facebook's platform, a 7.18% increase year-over-year.

- India has the most Facebook users with over 385.65 million
- US (188.6 million)
- Indonesia (136.35 million)
- Brazil (111.75 million)
- Mexico (94.8 million)
- …

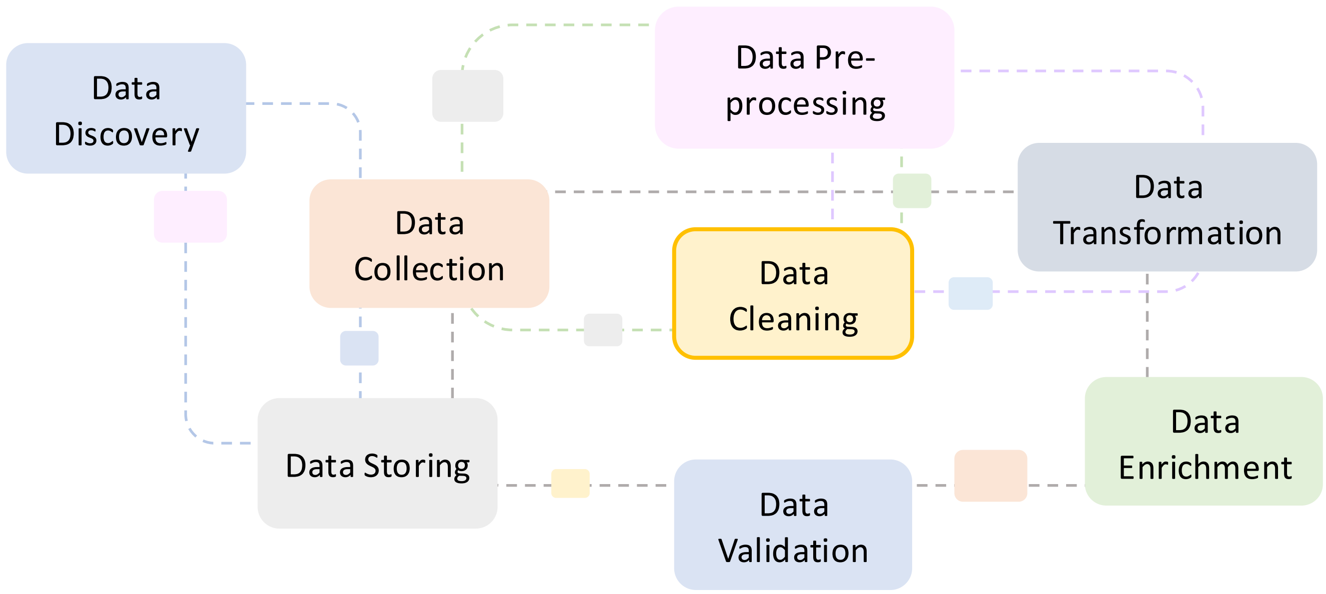


Brian Dean, Facebook User & Growth Statistics, Backlinko, https://backlinko.com/facebook-users

Data Pre-Processing

- Selection
- **Formatting**
  - Date formatting
    - DD/MM/YYYY
    - MM/DD/YYYY
    - YYYY/MM/DD
    - …
  - Numeric formatting
    - Decimal points
    - Thousands separators
  - Text formatting
    - Upper, lower, title case, whitespace, symbols
  - Categorical data formatting
    - ‘Male’ vs ‘M’; ‘Female’ vs ‘F’
  - Currency conversion
    - USD, AUD, EUR, CNY…
  - Unit conversion
    - Miles to kilometres
    - Fahrenheit to Celsius
  - Text encoding standardisation
    - UTF-8
  - Time zone alignment
  - Address formatting
  - File format conversion
    - XML → CSV
    - JSON → XML

Overview of Data Wrangling Process



Data Cleaning:

A critical step focused on removing or correcting inaccuracies, inconsistencies, duplicates, or irrelevant data points within datasets.

This step ensures the quality and reliability of data for analysis.

Data wrangling is the process of making data useful.

Data Cleaning

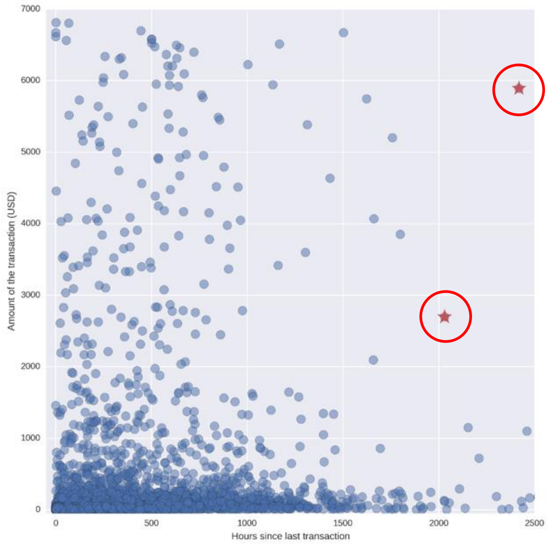
- Data anomalies
  - Items within a dataset that deviate from a typical pattern or expectation.
  - They can indicate errors.
  - They can also reveal insights or new patterns.

Missing Values

```
32,1,1,95,0,?,0,127,0,.7,1,?,?,1
34,1,4,115,0,?,?,154,0,.2,1,?,?,1
35,1,4,?,0,?,0,130,1,?,?,?,7,3
36,1,4,110,0,?,0,125,1,1,2,?,6,1
38,0,4,105,0,?,0,166,0,2.8,1,?,?,2
38,0,4,110,0,0,0,156,0,0,2,?,3,1
38,1,3,100,0,?,0,179,0,-1.1,1,?,?,0
38,1,3,115,0,0,0,128,1,0,2,?,7,1
38,1,4,135,0,?,0,150,0,0,?,?,3,2
38,1,4,150,0,?,0,120,1,?,?,?,3,1
40,1,4,95,0,?,1,144,0,0,1,?,?,2
```

The Switzerland heart disease data set from
[UCI machine learning data repository]

Outliers



Data Cleaning

- Data anomalies
  - Items within a dataset that deviate from a typical pattern or expectation.
  - They can indicate errors.
  - They can also reveal insights or new patterns.

Duplicate Data

Air France Flight 447	**Airbus A330-203**	Extremely High	Bad Visibility by Night	Sao Pedro	12	2009-06-01
American Airlines Flight 11	**Boeing 767-223ER**	Extremely High	Good Visibility by Day	US	11	2001-09-11
American Airlines Flight 587	**Airbus A300B4-605R**	Extremely High	Bad Visibility by Night	US	9	2001-11-12
American Airlines Flight 77	**Boeing 757-223**	Extremely High	Bad Visibility by Day	US	6	2001-09-11
Caspian Airlines Flight 7908	**Tupolev Tu-154M**	Extremely High	Bad Visibility by Night	Iran	12	2009-07-15
American Airlines Flight 77	**Boeing 757-223**	Extremely High	Bad Visibility by Day	US	6	2001-09-11
Iranian Air Force (15-2280)	**Ilyushin Il-76MD**	Extremely High	Bad Visibility by Night	Iran	18	2003-02-19
Kenya Airways Flight 431	**Airbus A310-304**	Extremely High	Bad Visibility by Night	Cote d'Ivoire	10	2000-01-30
Pulkovo Flight 612	**Tupolev Tu-154M**	Extremely High	Bad Visibility by Night	Ukraine	10	2006-08-22

Consistency Issue

Employer Database

Name	Age	Position	Salary
John Smith	30	Sales	2000
Kate Williams	23	Reception	1000

Staff Database

Name	Age	Position	Salary
John Smith	30	Sales	2500
Jane Jones	36		3000

Name	Age	Position	Salary
John Smith	30	Sales	2000
Kate Williams	23	Reception	1000
John Smith	30	Sales	2500
Jane Jones	36		3000

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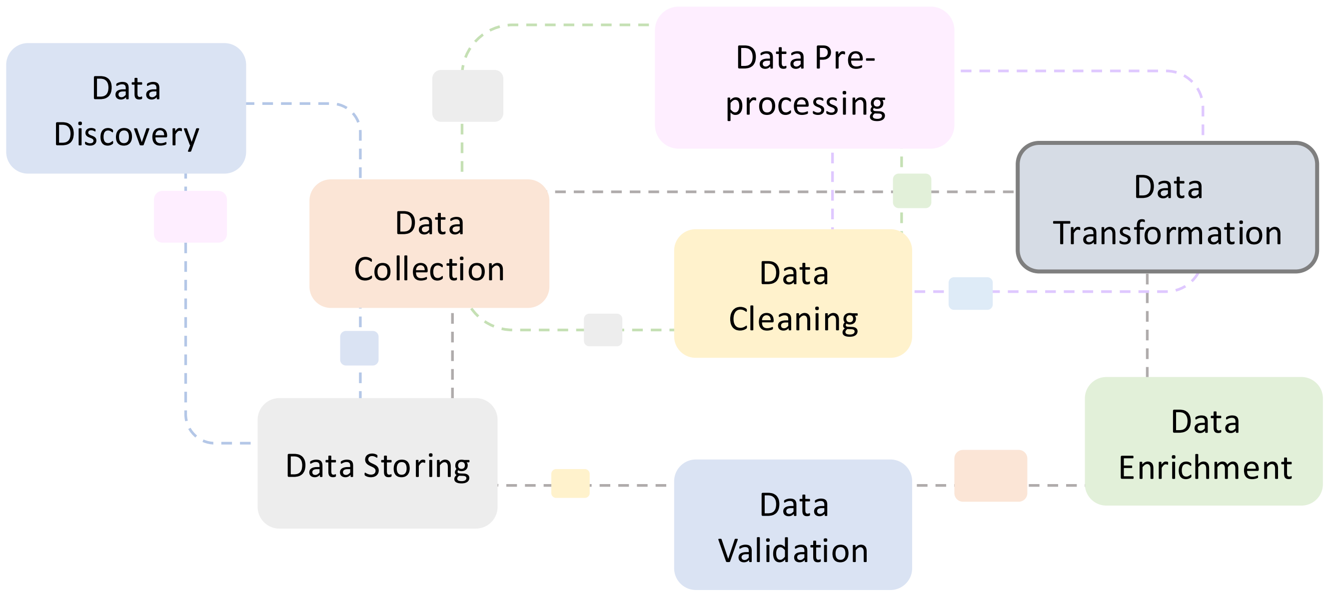
Overview of Data Wrangling Process



Data Transformation

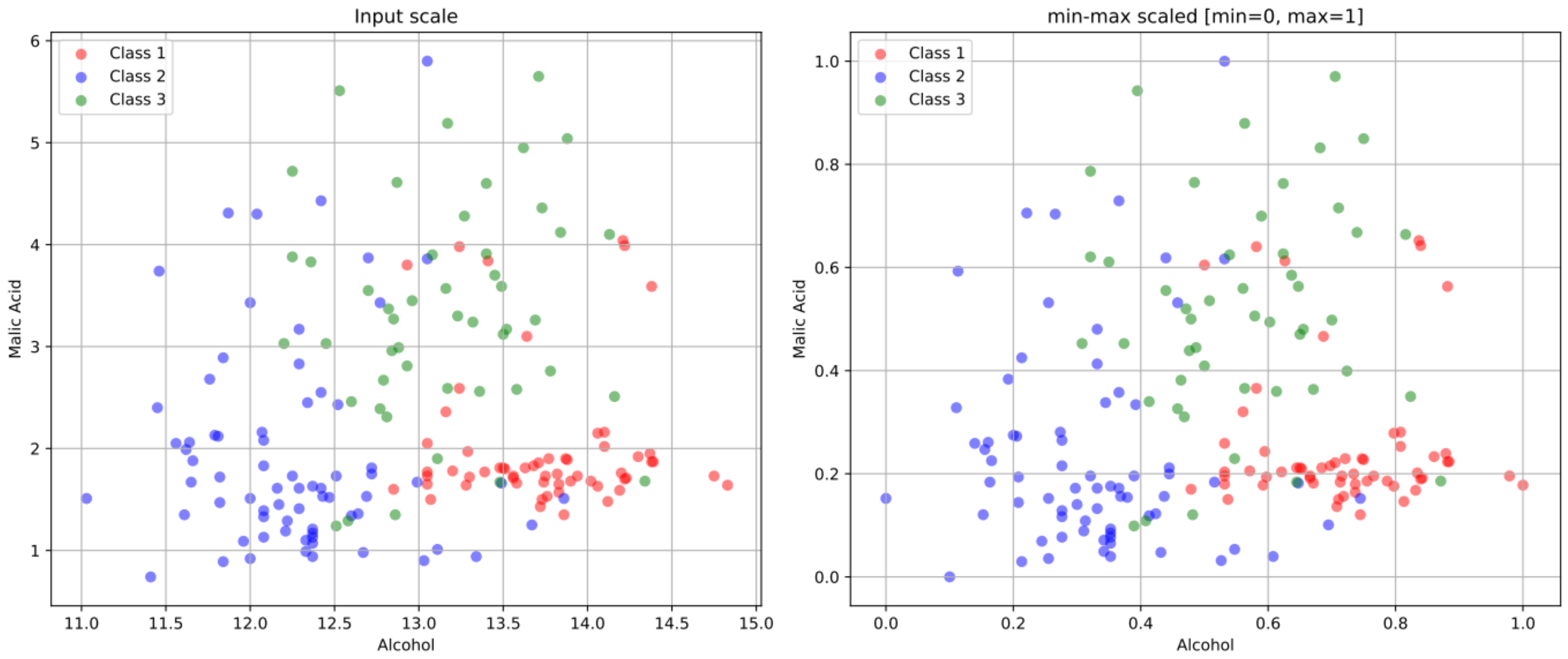
This involves converting data from its original form into a format or structure that is more appropriate for analysis.

Transformation can include normalization, aggregation, and feature engineering to enhance data analysis.

Data wrangling is the process of making data useful.

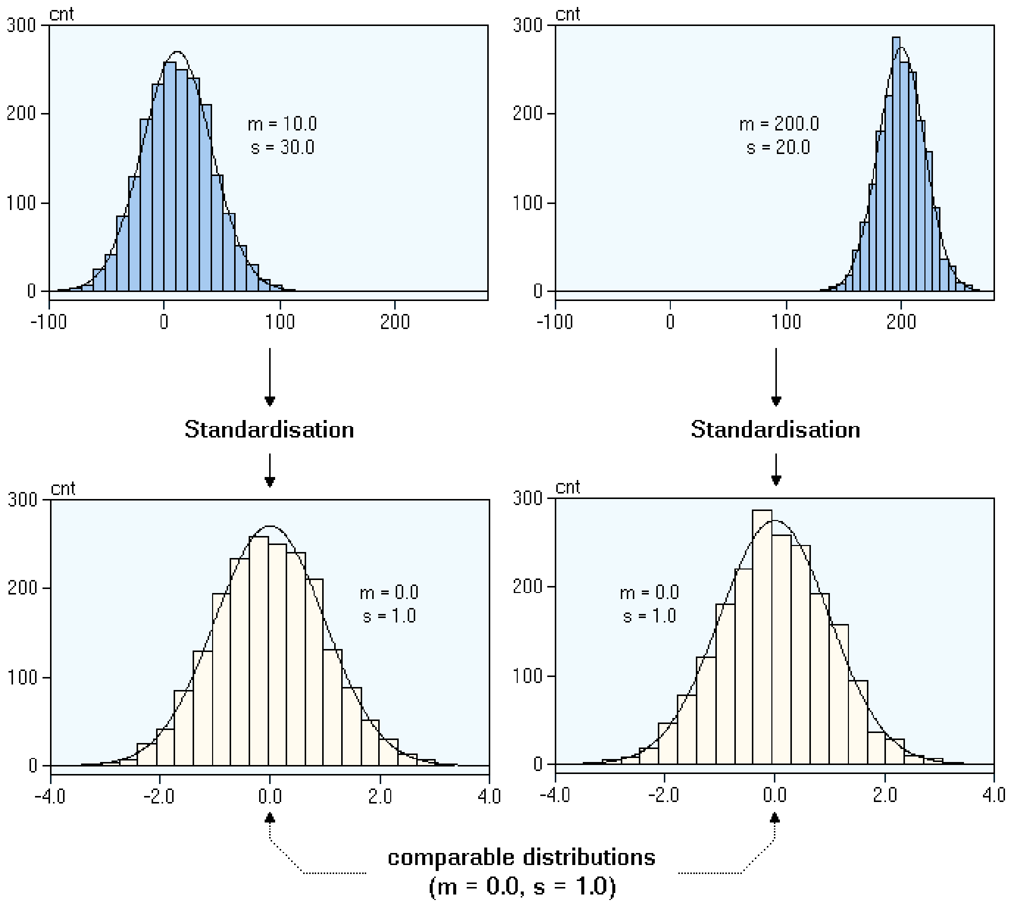
Data Transformation

- **Numeric data**
  - **Normalisation**
    - Rescales numeric data to fall within a specified range, often between 0 and 1, to ensure that no variable dominates due to its scale.



Data Transformation (cont.)

- **Numeric data**
  - Normalisation
  - **Standardization**
    - Adjusts the distribution of data so that it has a mean of 0 and a standard deviation of 1. This is useful for algorithms that assume data is normally distributed.



Data Transformation (cont.)

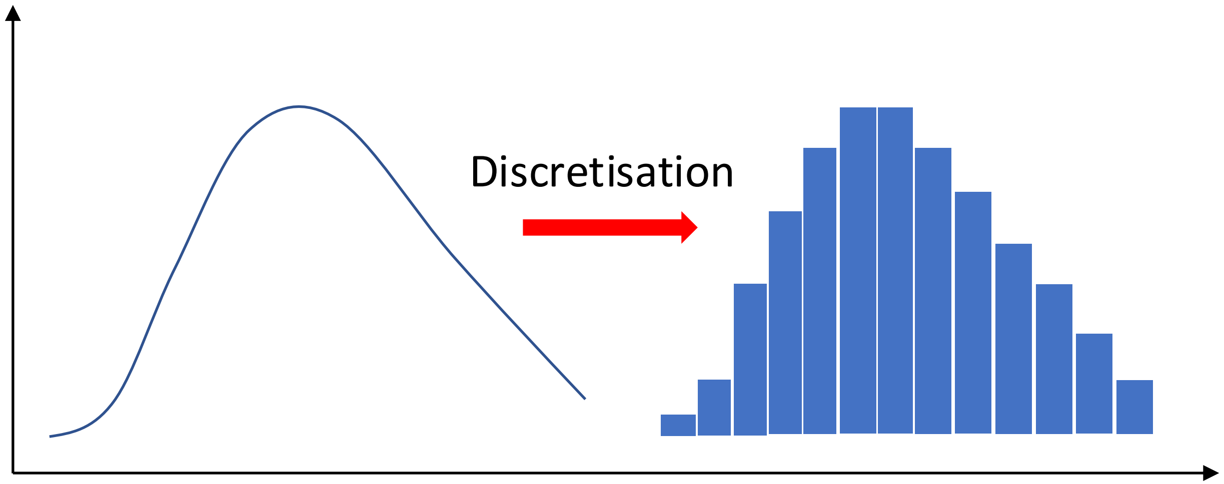
- **Numeric data**
  - Normalisation
  - Standardization
  - **Aggregation**
    - Summarizing or aggregating data, such as computing summary statistics (mean, median, sum) for groups of data. This can be particularly useful in reducing data dimensionality and extracting insights from the data.

	split	Casualties	Circumstances	Country	Crew	Date	Ground	Latitude	Location	Longitude	Notes	Passengers	Phase	Reason	Total Dead	Type
extract	extract1															
British Airways Flight 476 and	Hawker Siddeley Trident 3B and	Extremely High	Bad Visibility by Night	Croatia	14	1976-09-10	1	45.8839523	Vrbovec - Croatia	16.416766	No survivors	162	ENR	Accident	177	COM
China Airlines Flight 676	Airbus A300-622R	Extremely High	Bad Visibility by Night	China	14	1998-02-16	7	25.0492632	Dayuan - Taiwan - China	121.193945	No survivors	182	APR	Accident	202	COM
China Northwest Airlines Flight 2303	Tupolev Tu-154M	Extremely High	Bad Visibility by Night	China	14	1984-06-06	0	34.341568	Xian - China	108.940175	No survivors	146	ENR	Accident	150	COM
Nigeria Airways Flight 2120	Douglas DC-8-61	Extremely High	Bad Visibility by Night	Saudi Arabia	14	1991-07-11	0	21.5433333	Jeddah - Saudi Arabia	39.1727778	No survivors	247	APR	Accident	261	COM
Saudia Flight 163	Lockheed L-1011-200 TriStar	Extremely High	Bad Visibility by Night	Saudi Arabia	14	1980-08-19	0	24.7116667	Riyadh - Saudi Arabia	46.7241667	No survivors	287	ENR	Accident	301	COM
Swissair Flight 111	McDonnell Douglas MD-11	Extremely High	Bad Visibility by Day	Canada	14	1998-09-02	0	44.4927778	Atlantic Ocean - Peggys Cove - N.S. - Canada	-63.9175	No survivors	215	ENR	Accident	229	COM
Union de Transportes Aériens Flight 772	McDonnell Douglas DC-10-30	Extremely High	Bad Visibility by Night	Niger	14	1989-19-09	0	18.9546137	Tenere - Niger	10.9134609	No survivors	156	ENR	Attack	170	INB

split	Crew	Ground	Latitude	Longitude	Passengers	Total Dead
count	58.000000	58.000000	58.000000	58.000000	58.000000	58.000000
mean	15.396552	51.017241	25.337876	21.807698	200.724138	257.155172
std	22.838825	240.369561	28.037558	74.024614	87.867283	232.558071
min	0.000000	0.000000	-100.423368	-100.423368	0.000000	0.000000
25%	9.000000	0.000000	13.835361	-52.617466	159.000000	171.750000
50%	12.000000	0.000000	30.489219	15.865327	181.000000	195.000000
75%	15.000000	0.000000	41.964744	76.751146	241.500000	263.250000
max	181.000000	1600.000000	55.755826	167.075701	560.000000	1692.000000

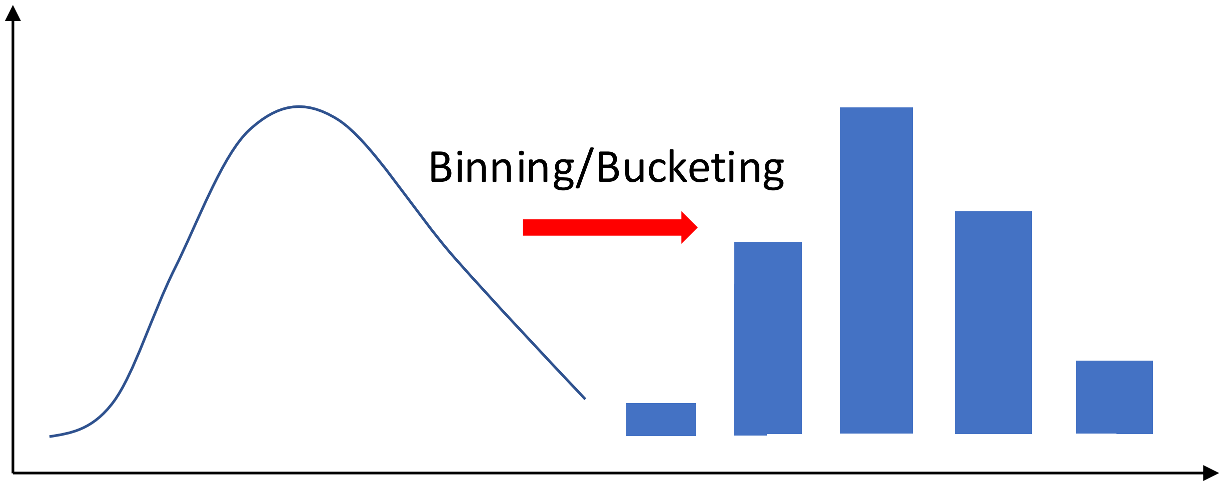
Data Transformation (cont.)

- **Numeric data**
  - Normalisation
  - Standardization
  - Aggregation
  - **Discretisation**
    - Converts continuous variables into discrete ones by creating a set number of equal-width or equal-frequency intervals, which can be useful for certain types of models.



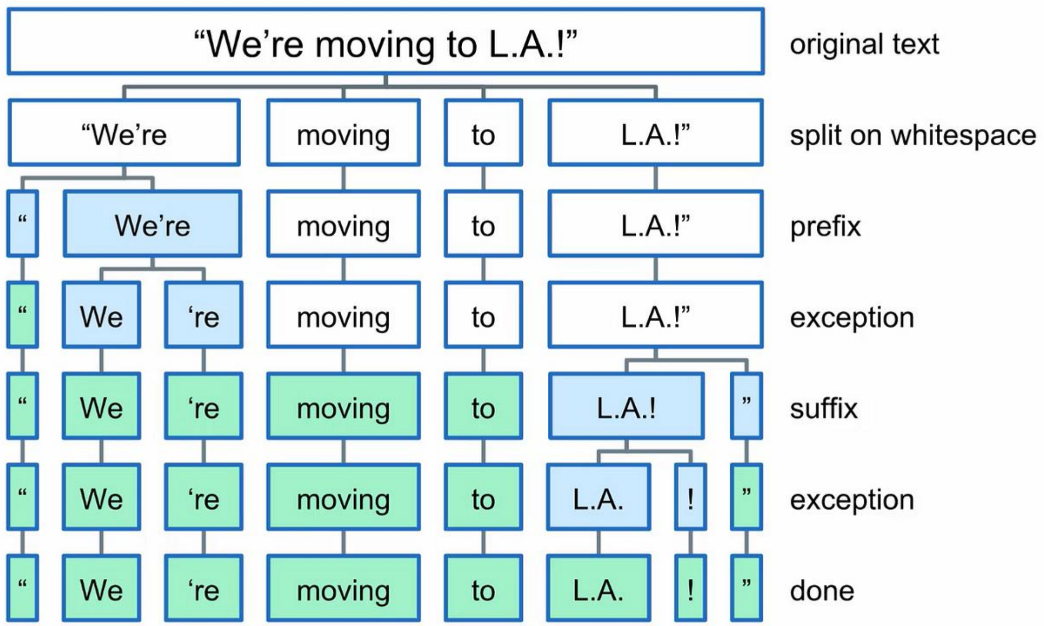
Data Transformation (cont.)

- **Numeric data**
  - Normalisation
  - Standardization
  - Aggregation
  - Discretisation
  - **Binning/Bucketing**
    - Converts numeric variables into categorical counterparts by grouping values into bins or categories, which can simplify the model and help in identifying trends.



Data Transformation (cont.)

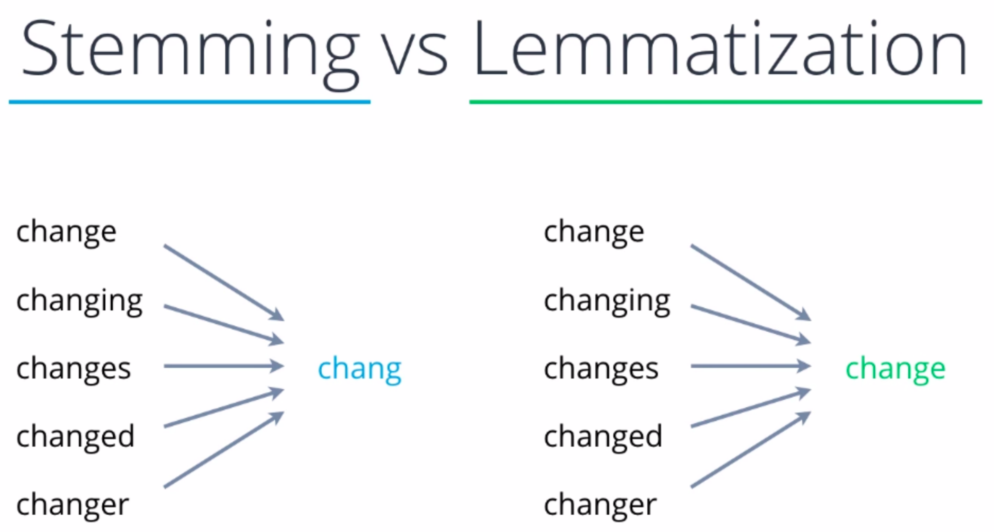
- **Text data**
  - **Tokenisation**
    - Breaking text into individual words or phrases



Utkarsh Kant, Tokenization – A complete guide, Medium, https://medium.com/@utkarsh.kant/tokenization-a-complete-guide-3f2dd56c0682

Data Transformation (cont.)

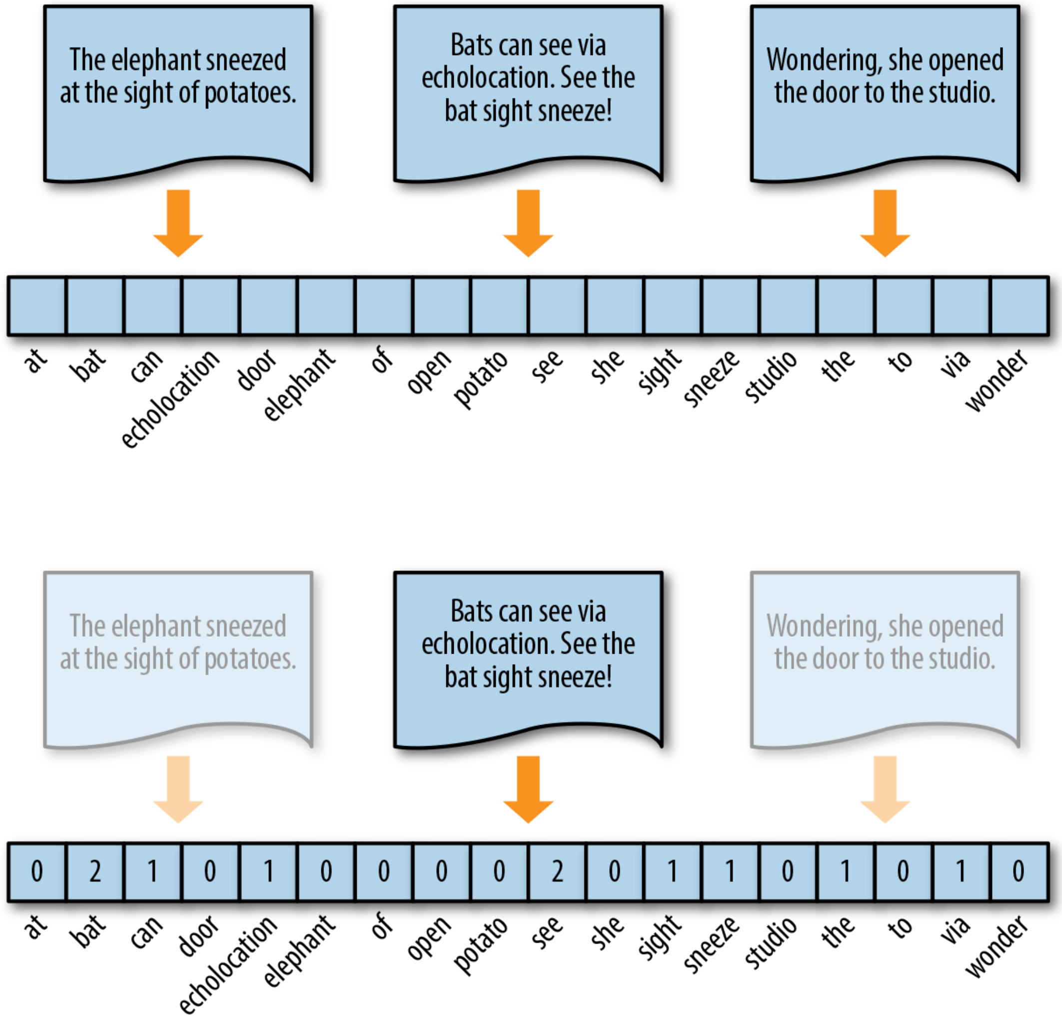
- **Text data**
  - Tokenisation
  - **Stemming/lemmatisation**
    - Reducing words to their root form



Botpenguin, Stemming, https://botpenguin.com/glossary/stemming

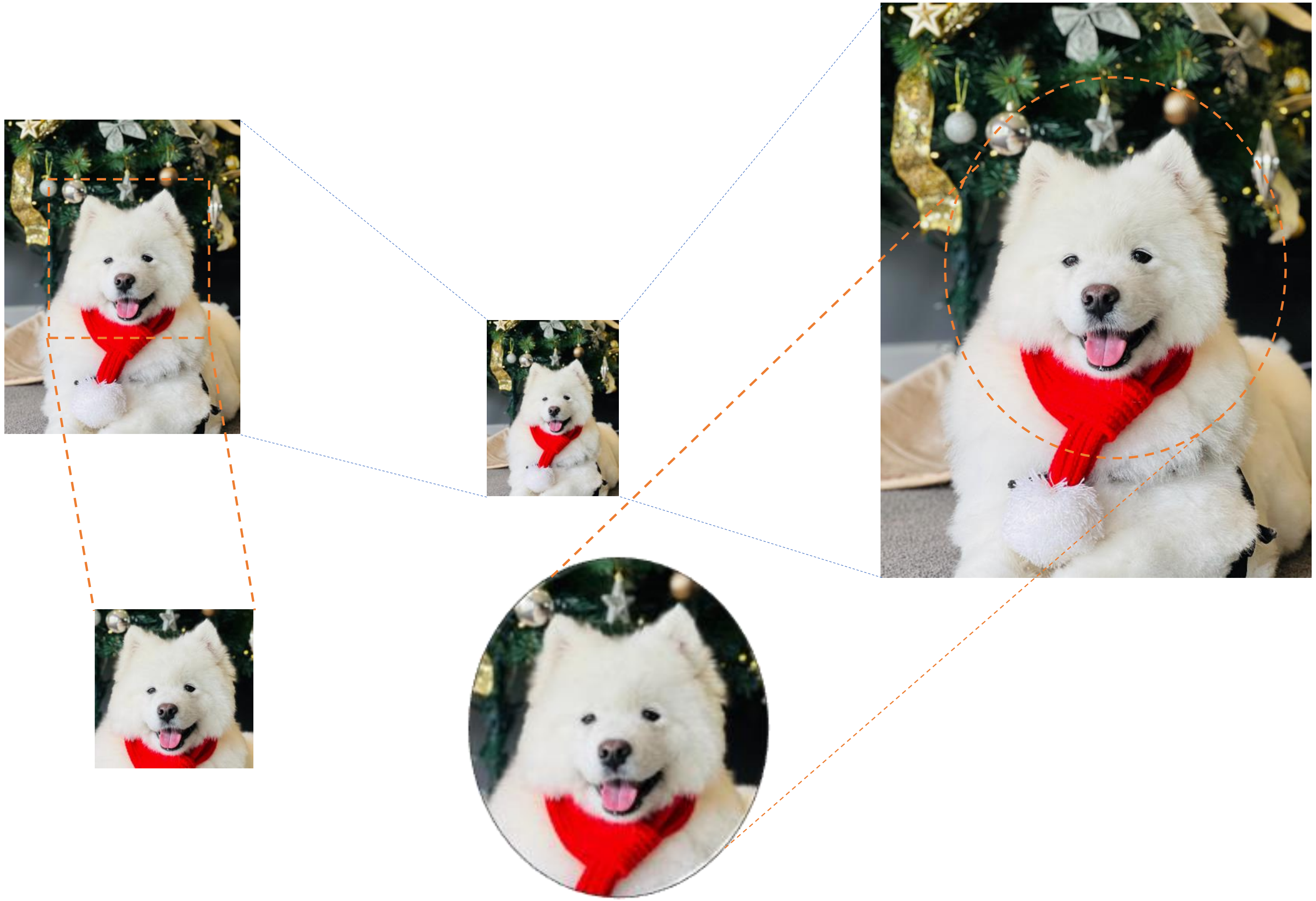
Data Transformation (cont.)

- **Text data**
  - Tokenisation
  - Stemming/lemmatisation
  - **Vectorisation**
    - Converting text into numerical format for analysis



Data Transformation (cont.)

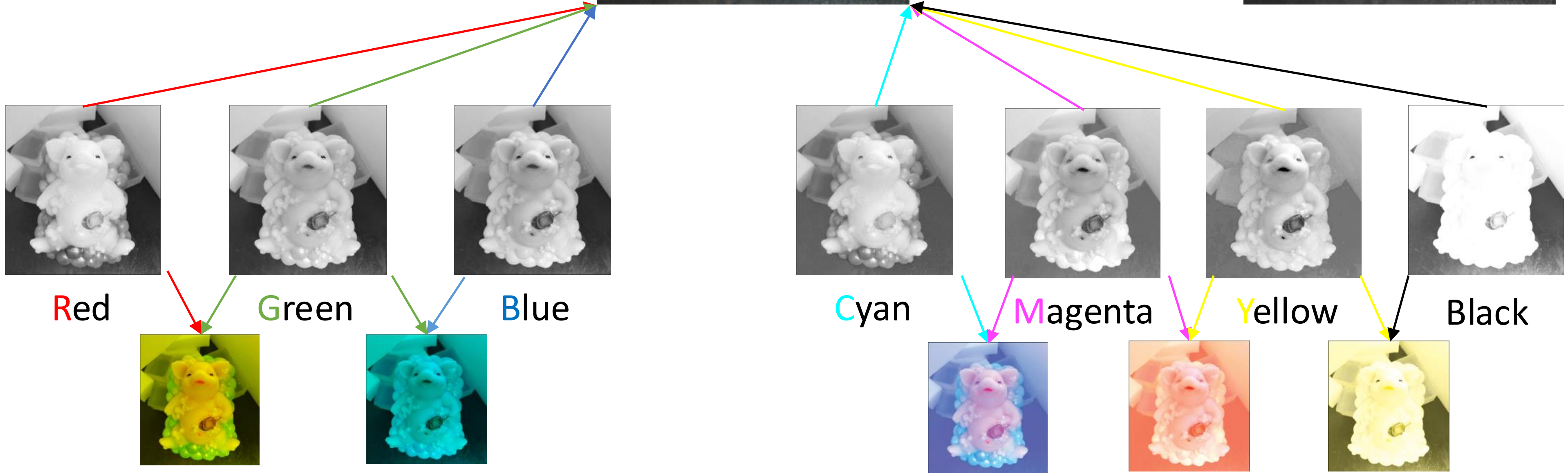
- Image data
  - Resize
  - Chopping



MONASH University

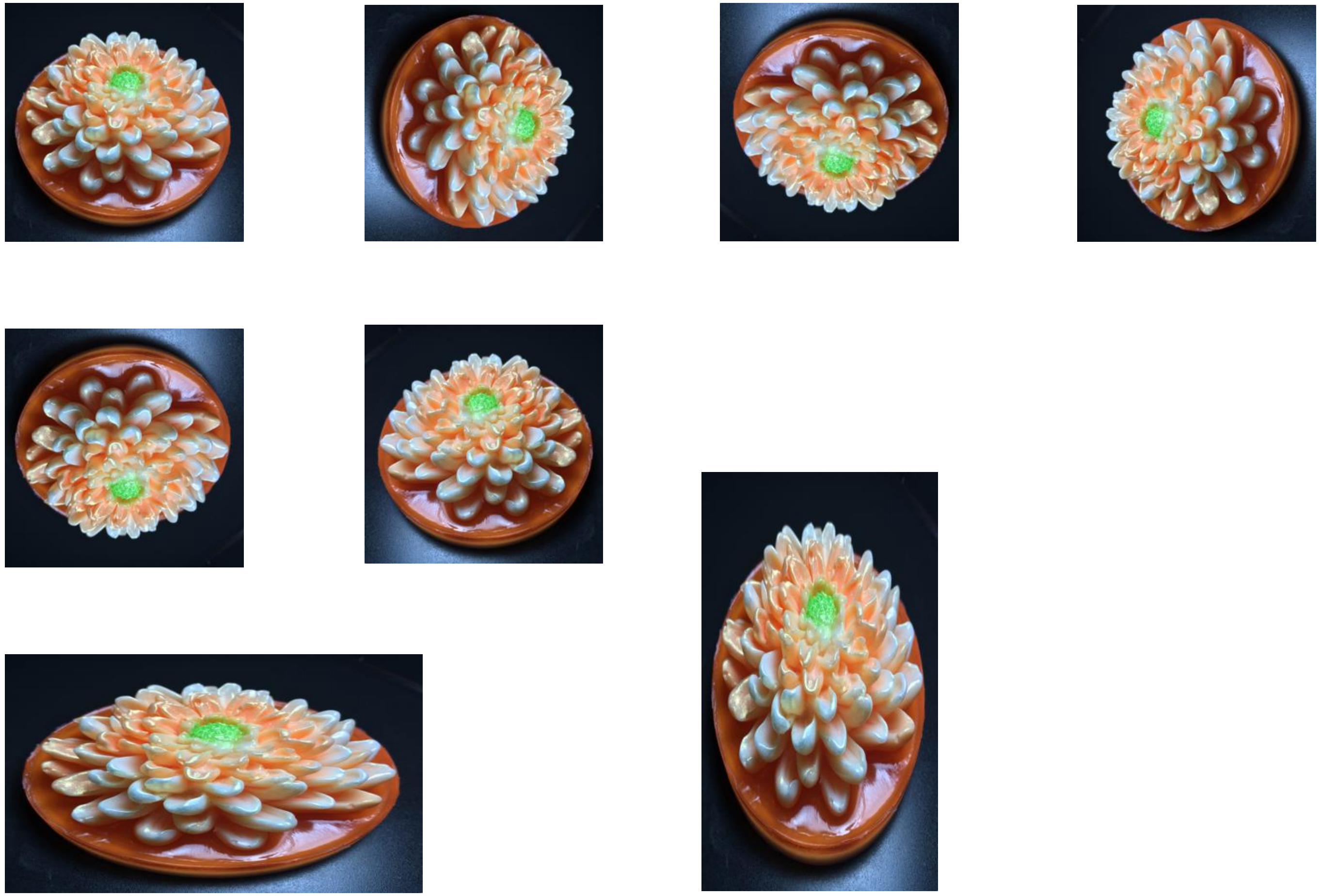
Data Transformation (cont.)

- **Image data**
  - Resize
  - Chopping
  - **Normalization**
  - **Colour space conversion**



Data Transformation (cont.)

- **Image data**
  - Resize
  - Chopping
  - Normalization
  - Colour space conversion
  - **Rotation**
  - **Flipping**
  - **Translation**



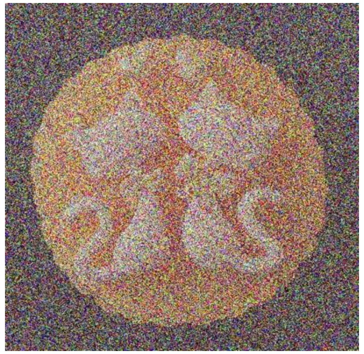
Data Transformation (cont.)

- **Image data**
  - Resize
  - Chopping
  - Normalization
  - Colour space conversion
  - Rotation
  - Flipping
  - Translation
  - **Noise injection**
  - **Brightness adjustment**
  - **Contrast adjustment**

Brightness +150

Brightness -150

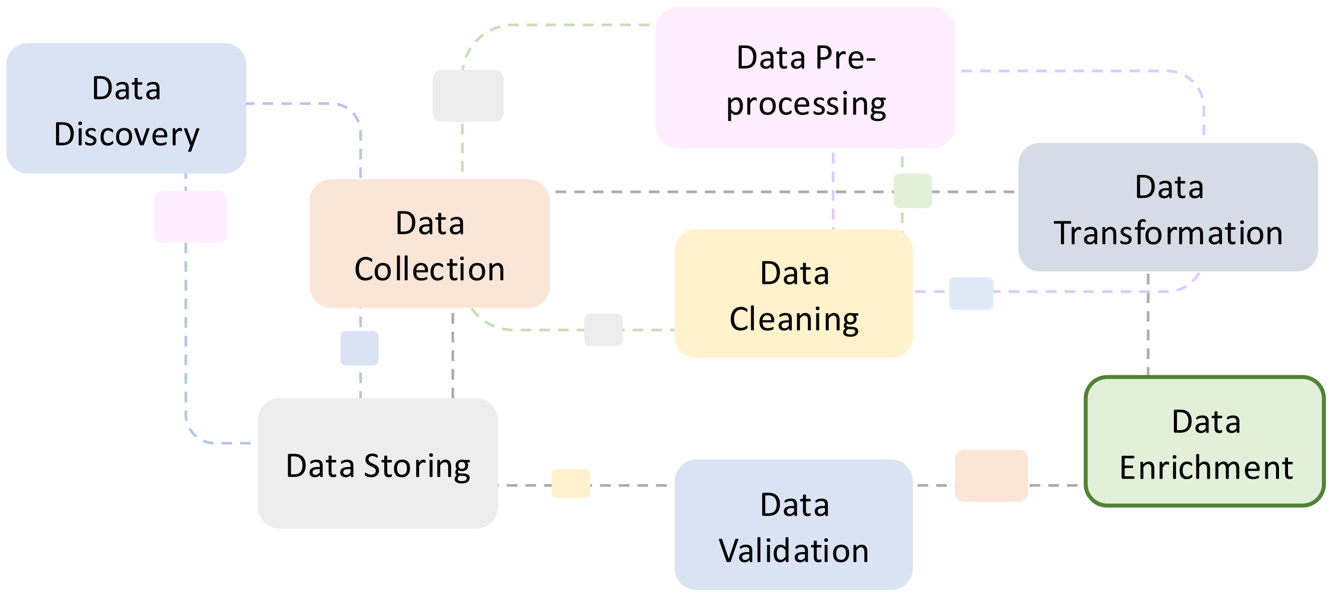
Noise



Contrast +150

Contrast -150

Overview of Data Wrangling Process



Data Enrichment

Enhancing the data with additional context or merging it with other data sources to make it completer and more informative.

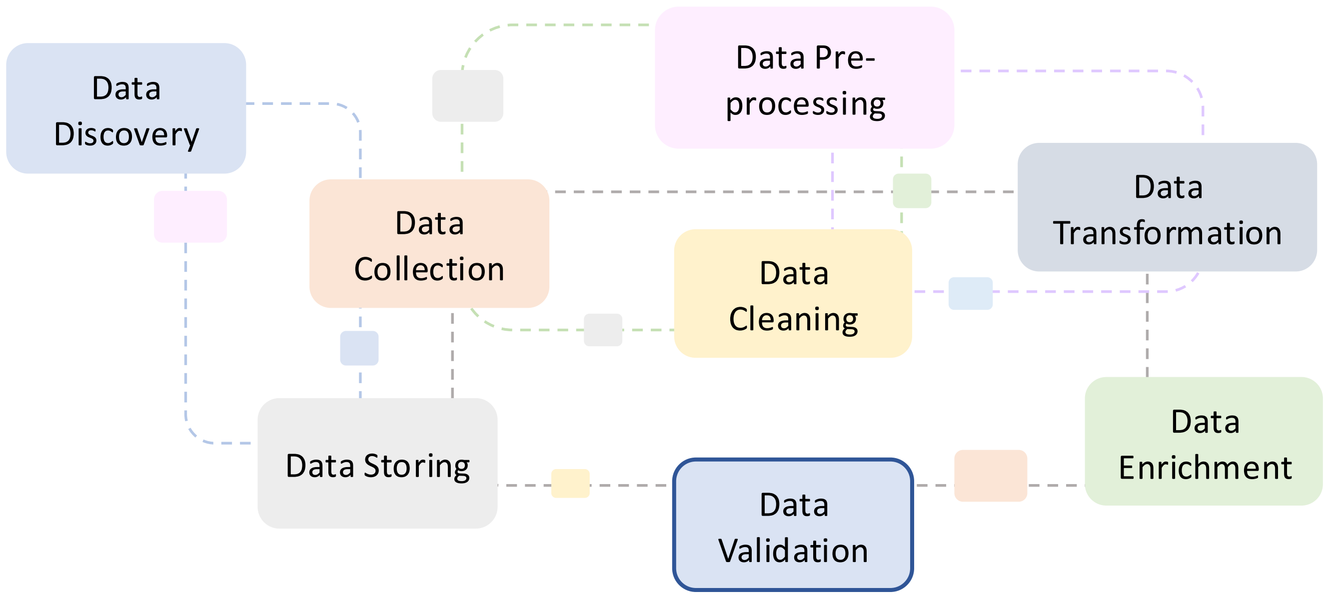
This step can involve adding demographic information, geographical tags, or combining datasets to provide more depth.

Data wrangling is the process of making data useful.

Data Enrichment

- **Data integration**
  - Combining data from different sources to create a more comprehensive dataset.
- **Data augmentation**
  - Adding information to existing records to enhance the depth of data on each data point.
- **Attribute enrichment**
  - Enhancing existing datasets by adding new attributes or features that were not previously included.
- **Temporal enrichment**
  - Adding time-related data to datasets.
- **Semantic enrichment**
  - Adding metadata or other semantic information to make data more understandable and usable.

Overview of Data Wrangling Process



Data Validation

Ensuring that the data is accurate, consistent, and usable for analysis.

Validation checks are performed to verify data quality and correctness after cleaning and transformation.

Data wrangling is the process of making data useful.

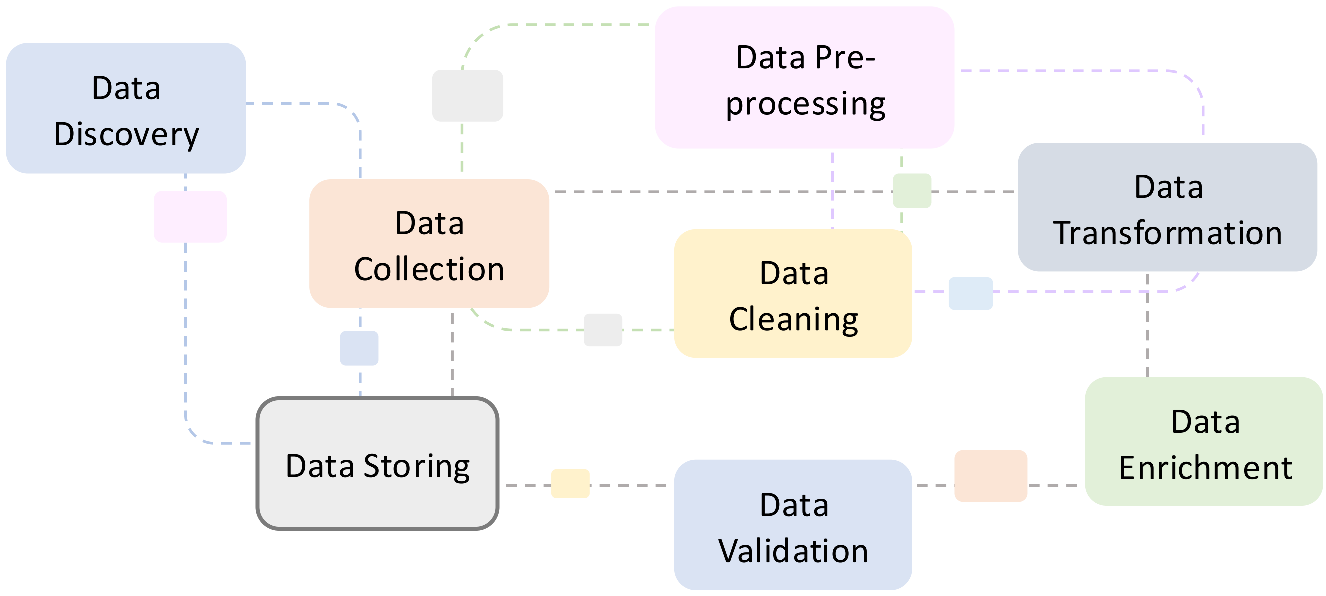
Data Validation

- **Define Validation Rules and Criteria**
  - Establish the standards that your data must meet.
- **Check for accuracy**
  - Ensure that the data accurately reflects the real-world entities or values it represents.
- **Ensure consistency**
  - Verify that the data is consistent within the dataset and across related datasets.
- **Validate data completeness**
  - Ensure no critical data is missing, which could impact analysis.
- **Test for logical integrity**
  - Confirm that the data makes logical sense and adheres to known constraints or relationships.
- **Validate range and constraints**
  - Ensure that data values fall within acceptable ranges or constraints.

Data Validation (cont.)

- **Format validation**
  - Verify that the data is in the expected format or structure, which is essential for automated processing.
- **Uniqueness checks**
  - Ensure that entries that are supposed to be unique do not have duplicates.
- **Cross-validation**
  - Use related datasets or data sources to validate the data.
- **Automate validation processes**
  - Streamline validation to make it efficient and repeatable, especially for large datasets or ongoing data collection.
- **Document validation issues and resolutions**
  - Keep track of identified issues and how they were resolved for future reference and accountability.

Overview of Data Wrangling Process



Data Storing

The final step involves saving the wrangled data in a suitable storage system for easy access and analysis.

This could be databases, data warehouses, or cloud storage solutions, depending on the scale and purpose of the data analysis project.

Data wrangling is the process of making data useful.

Data Storing

- **Selection of Storage Solution**
  - Choose an appropriate data storage solution that aligns with the data's nature, size, and intended use.
- **Data Modelling and Schema Design**
  - Design a logical structure for the data that supports efficient storage, query, and retrieval.
- **Normalization and Denormalization**
  - Organize the data to reduce redundancy and improve integrity in relational databases through normalization, or optimize performance and read efficiency through denormalization, especially in NoSQL databases or data warehouses.
- **Data Formatting and Encoding**
  - Ensure that data is stored in a consistent and appropriate format that matches the storage system's requirements.
- **Implementing Data Security Measures**
  - Protect data from unauthorized access and ensure privacy and compliance with data protection regulations (e.g., GDPR, HIPAA).

Data Storing (cont.)

- **Data Indexing and Optimization**
  - Enhance the speed and efficiency of data retrieval.
- **Backup and Recovery Planning**
  - Ensure data durability and the ability to recover from data loss or corruption.
- **Data Lifecycle Management**
  - Manage the lifecycle of data from creation to deletion, aligning with data retention policies and legal requirements.
- **Documentation and Metadata Management**
  - Provide context and understanding for the data and how it's stored, facilitating easier data management, governance, and use.
- **Monitoring and Maintenance**
  - Ensure the storage system remains efficient, scalable, and reliable as data volume grows and access patterns change.

Summary & To-do List

- Please download and read the materials provided on Moodle.
- Watch the video introducing Pandas basic functions.

- Assessments
  - Group Self-Selection is open - due on 30 March, Monday
  - Group Assessment 1 will be released this week.

- Next week: Regular Expression (Challenging topic!!!)